CarbonLedger — Scope 3 Carbon Accounting and Disclosure Agent

Track: Sustainability & Social Impact

CarbonLedger compresses the cost of producing a Scope 3 report from "$30,000 + 3 weeks" to "1 hour + near-zero marginal cost", while making every single number traceable back to its source document.

Live demo: https://myproject-dc19d.web.app

Anonymous sign-in · No registration

Evidence is scattered. Expertise is expensive.

01

Scattered evidence.

Invoices, scans, photos and spreadsheets live with different suppliers. Manual compilation takes 2–3 weeks per report.

02

Semantic, not string match.

"Air-freighting 1,200 kg of temperature-sensitive goods from Long Beach to Tokyo" is a reasoning task — never a keyword lookup.

03

A bill that repeats.

Consultant reports cost $20,000–$50,000, are re-done every year, and stay hard for auditors to verify.

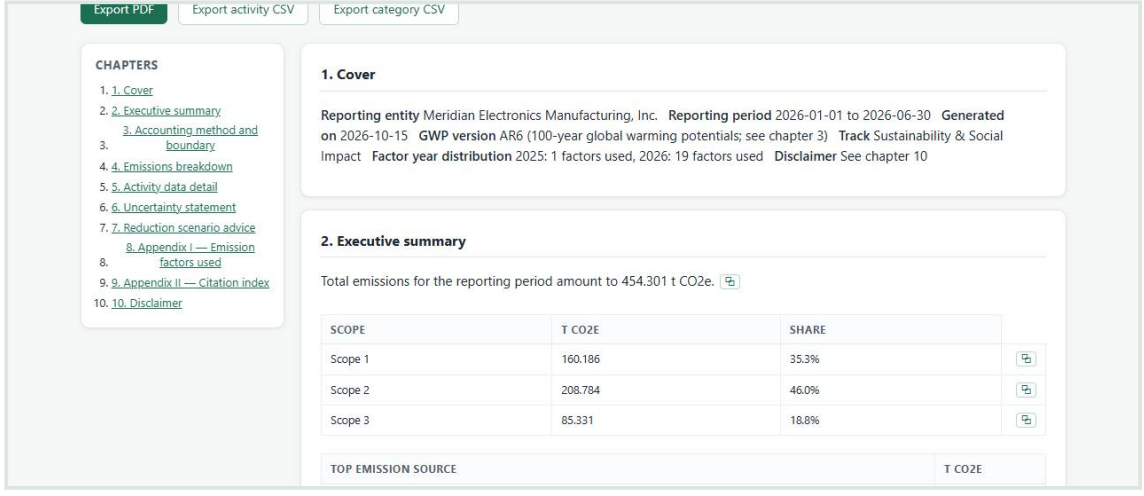
2–3 WEEKS
of manual work per report

$20,000–$50,000
consultant fee, every year

HARD TO VERIFY
for auditors, line by line

One upload. A full, traceable report.

- **Mixed upload.** PDF including scans, photos (JPG/PNG), XLSX and CSV — one task, one queue.
- **Classify + extract.** Every page classified; activity data, quantities, units and locations extracted.
- **Match + grade.** Public factors matched semantically; uncertainty graded per record.
- **Compute.** Scope 1/2/3 totals, three reduction scenarios, GWP folding with AR5/AR6 disclosure.
- **Disclose.** A ten-chapter GHG Protocol report with PDF/CSV export.
- **Trace everything.** Each number carries a citation anchor that jumps back to its source document.

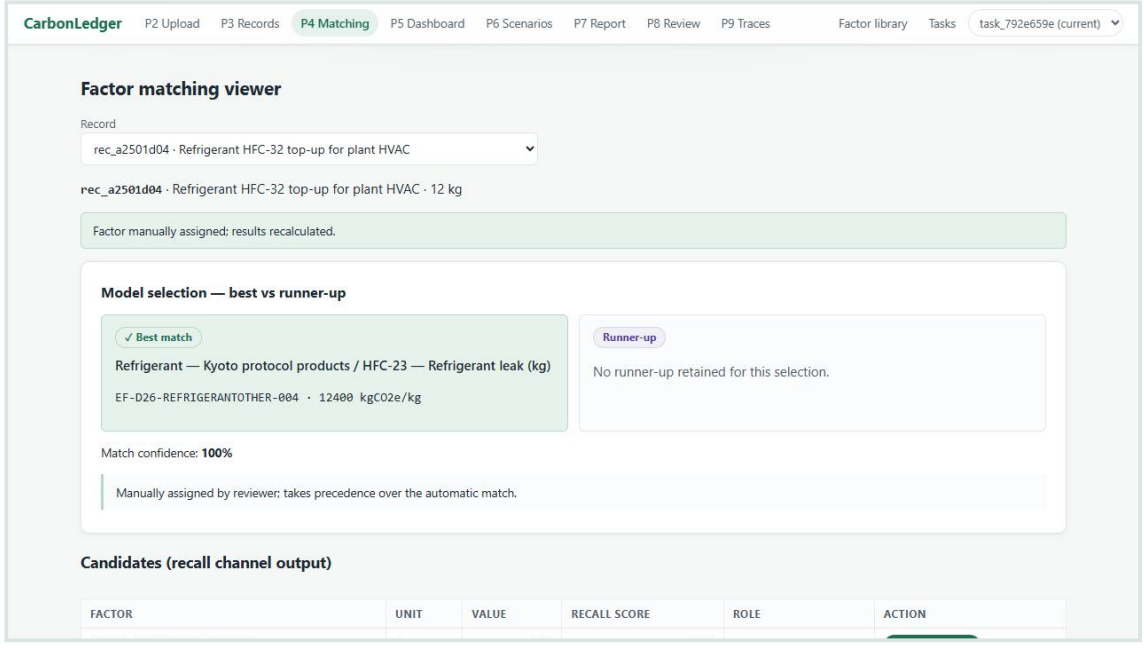


Live UI: ten-chapter report reader with clickable citation anchors.

Factor matching is a reasoning problem.

- **1 · Recall.** Dual-channel recall: keyword plus official embeddings (gemini-embedding-001); 5–15 candidates with threshold relaxation.
- **2 · Select.** The model writes a 40+ word comparative rationale and retains the runner-up for review.
- **3 · Grade.** Rule-based uncertainty grading with unit-caliber self-checks auditors can recompute.

Remove the model and the system cannot run: rule engines cannot choose between boundary calibers. This is the differentiator — not a wrapper.



Live UI: best-vs-runner-up comparison with the stored rationale and recall scores.

All generative calls: gemini-3.5-flash-lite, structured output.

Trust is engineered, not promised.

Structured output everywhere

Every generative call uses a response schema. No free-text parsing anywhere in the codebase.

Closed enums, validated

Enum violations are retried with the error fed back — never silently accepted.

Candidates only

A factor id outside the candidate list is a failed call. It is retried, never used.

Citation veto

Every numeric conclusion carries an anchor. One dangling reference vetoes the whole verdict.

Measured: 100% citation coverage · 0 invalid references

From upload to audited disclosure.

Eleven steps, fully traced

1 task creation

2 upload & parsing

3 page classification

4 field extraction

5 normalization

6 boundary decision

7 factor recall

8 factor selection

9 uncertainty grading

10 calculation

11 scenarios → report & review

Uncertain records route to a human review queue. Every AI call logs latency, tokens and cost into ai_traces.

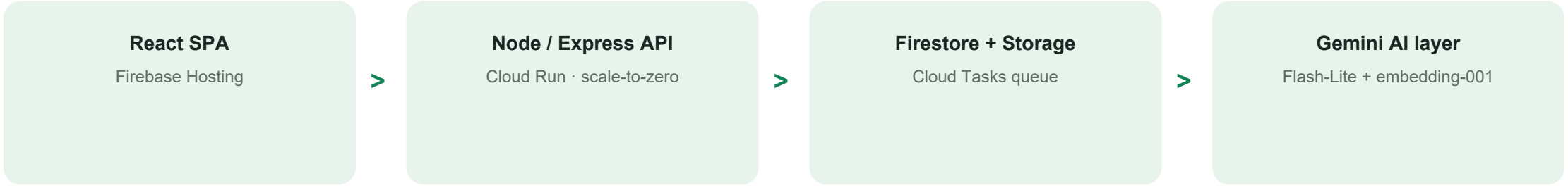
741

public emission factors, shipped offline

UK DESNZ 2026 + US EPA 2025 + USEEIO · five industries · loading never touches the network

Data-only growth: new industries need zero code change.

Boring infra. Serious guarantees.



- **Zero-barrier access.** Anonymous sign-in completes automatically; a judge uses the full flow with no registration. Per-uid task ownership isolates every visitor.
- **Reproducible offline.** Local storage adapters run the identical code for tests and the offline evaluation tool — no mocks in measured metrics.
- **Observable.** Cloud Logging/Trace plus per-call ai_traces: input/output summary, latency, tokens and cost estimate.

Measured, not claimed — 11/11 PASS.

Full run on the real adapter and the public deployment. Overall verdict: PASS.

Metric	Target	Measured
Document classification (Top-1)	≥ 93%	**29/30 (96.7%)**
Field extraction accuracy	avg ≥ 95%; qty/unit ≥ 98%	**100.0% (qty 100%, unit 100%)**
Factor matching (Top-1 / Top-3)	≥ 85% / ≥ 95%	**34/40 (85.0%) / 39/40 (97.5%)**
Scope decision accuracy	≥ 90%	**scope 9/10, category 12/12**
Calculation correctness	100%	**10/10 exact**
Citation coverage	100%	**100%**
Invalid references (veto)	0	**0**
End-to-end P90, 20 docs	≤ 3 min	**61.3 s**
AI cost per document	≤ $0.03	**$0.00058**

Machine-readable JSON + human-readable markdown: eval/results/run-online/. P90 latency from 10 public one-click runs; cost covers 102 model calls per 20-document run.

Every report ships with a reduction plan.

S-1 · Mode shift

Move freight to lower-carbon transport modes; savings computed from the same ledger math.

S-2 · Supplier swap

Re-point activity to cleaner suppliers or grids and see the delta instantly.

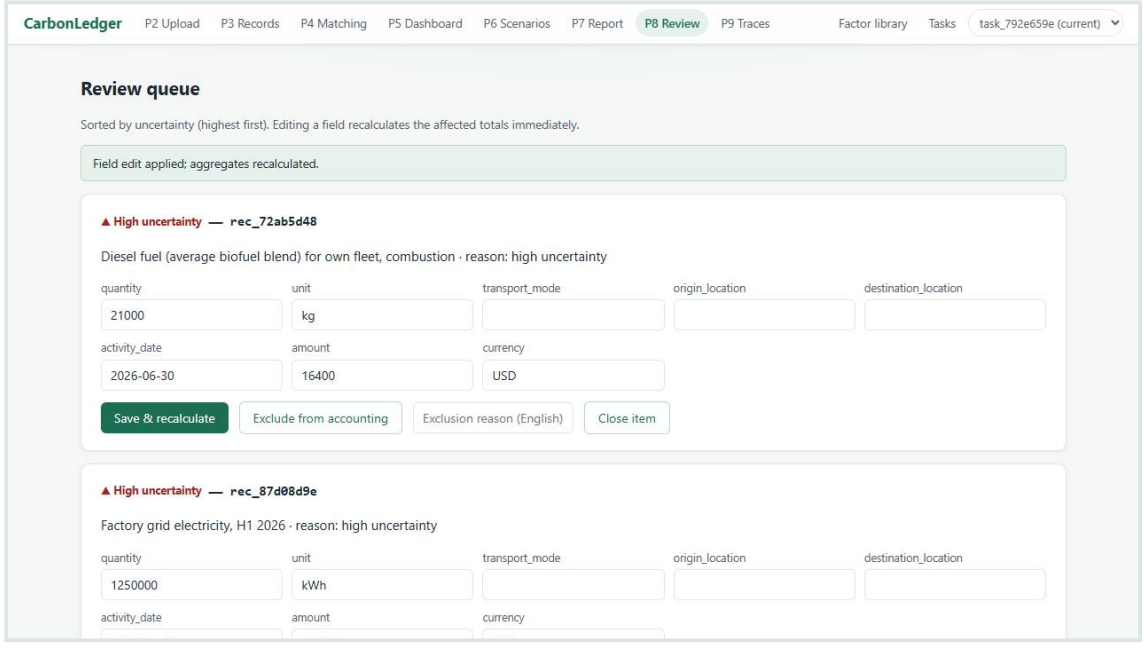
S-3 · Material substitution

Replace carbon-intensive inputs; every figure stays hand-recomputable.

Adjustable ratios for what-if planning. Scenarios with no applicable activity are stated honestly as zero — never padded to look good.

Humans decide where it matters.

- **Sorted by doubt.** The review queue ranks records by uncertainty, highest first — reviewers start where risk is.
- **Edits recalculate.** A field edit immediately recalculates affected totals and refreshes the report.
- **Overrides are labelled.** Manual factor assignments are tagged, take precedence, and stay auditable.
- **Exclusions need reasons.** Excluded records require an English reason and still appear in the report.

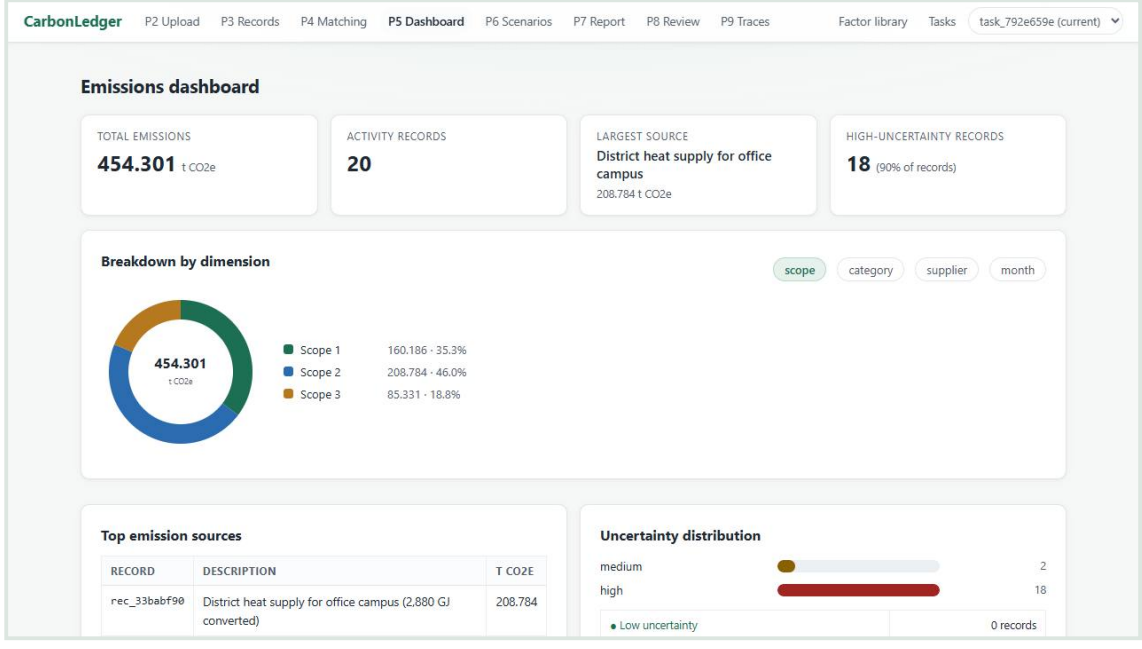


Live UI: review queue with inline editing, exclusion reasons and instant recalculation.

From a $30,000 engagement to self-serve.

	Consultant	CarbonLedger
Cost	$20,000–$50,000, yearly	**Near-zero marginal cost ($0.00058/doc)**
Time	2–3 weeks	**About 1 hour (P90 61.3 s / 20 docs)**
Trust	Hard for auditors to verify	**Every number traceable to source**
Reuse	A new bill each period	**Reusable every reporting period**

SMEs self-serve a compliance artifact they previously outsourced — then reuse it every period.



Live UI: emissions dashboard — scope donut, top sources, uncertainty split.

What comes next.

- **More industries, data-only.** New sectors arrive as factor-file additions — zero code change to the pipeline.
- **Supplier-specific factors.** Intake path for primary supplier data alongside public averages, with provenance kept.
- **Assurance-ready exports.** Evidence packs shaped for third-party verification: ledger, citations, method notes.
- **Why it scales.** The factor library is data; the pipeline is industry-agnostic; the review loop compounds trust.

Built for this track

Entered in AI Builder Cup 2026, Sustainability & Social Impact: measurable decarbonization leverage for every SME supplier in a value chain.

TRACK: SUSTAINABILITY & SOCIAL IMPACT

Every number traceable.
Every report reproducible.

Try it now — anonymous sign-in, one-click 20-document dataset, ten-chapter report in about a minute.

https://myproject-dc19d.web.app

Full source code, evaluation evidence and this deck ship with the submission. Thank you.